

ESG Intelligence Briefing — August 03, 2026

Executive Summary

Kazakhstan's construction and export sectors face immediate and substantial financial implications from the EU's Carbon Border Adjustment Mechanism (CBAM), effective 2026, particularly for steel and aluminum. Exporters failing to accurately report emissions risk significantly higher charges based on EU default values. Concurrently, Kazakhstan's new Construction Code, active since July 1, 2026, introduces new material consumption and cost accounting standards. The global construction industry continues its shift towards low-carbon materials, with tightened procurement rules and increased investor scrutiny pushing for verified carbon disclosures and performance-based concrete standards, which are critical for maintaining competitiveness and reducing CBAM liabilities.

What Changed Since Last Briefing

Previous briefings noted the looming impact of CBAM; this briefing solidifies the financial reality, detailing the cost implications for Kazakhstan's steel and aluminum exports if emissions are not accurately monitored and reported. We also have confirmed the effective date and key changes of Kazakhstan's new Construction Code. Regrettably, automated embodied carbon calculations (both factor lookup and LCA estimation) are currently unavailable due to tool limitations, preventing quantitative updates on material carbon footprints and project-level savings compared to previous runs where this data was also unavailable.

1. Regulation Watch

- **EU Carbon Border Adjustment Mechanism (CBAM) - Live Financial Obligation:**
 - **What it is:** The EU's CBAM became a live financial obligation on January 1, 2026, imposing carbon costs on imports of carbon-intensive goods, including iron, steel, cement, and aluminum, equivalent to the carbon price paid by EU producers.
 - **Who it affects:** Directly impacts Kazakhstan's aluminum, iron, and steel exporters to the EU. The EU accounted for 54% of Kazakhstan's aluminum exports.
 - **Effective date:** January 1, 2026 (financial obligation).
 - **Business implication:** Companies failing to monitor and report actual emissions risk being charged based on significantly higher EU default values, including additional markups of 10% in 2026, rising to 30% by 2028. This could lead to CBAM costs exceeding €108 million (US\$125 million) for iron and steel, potentially 123% of some export values. Properly monitored and reported emissions could halve this liability to around €57 million (US\$66 million) annually. Kazakhstan's low domestic carbon price (~€0.40/ton vs. EU's ~€80/ton) means exporters effectively pay the full EU carbon price. (Source: The Astana Times, June 4, 2026).
- **Kazakhstan Construction Code 2026 Enacted:**
 - **What it is:** The new Construction Code, signed January 9, 2026, replaces previous laws and brings significant changes to construction activities within Kazakhstan.
 - **Who it affects:** All construction companies, contractors, and project managers operating in Kazakhstan.
 - **Effective date:** July 1, 2026.
 - **Business implication:** Key changes include new internal material consumption standards and actual purchase price recording for construction costs. Businesses must update accounting policies and cost control procedures to comply with these new regulations. (Source: JDSupra, 2026).

2. Construction & Market News

- **Significant Investment in Kazakhstan's Construction Sector:** Kazakhstan's construction industry attracted over \$7 billion in investment in 2026, primarily for housing and urban renewal projects, signaling robust domestic demand. The government plans to build and repair over 11,000 km of roads in 2026, continuing work on major corridors like Kyzylorda–Zhezkazgan. (Source: The Astana Times, February 5, 2026; primeminister.kz, 2026). This ongoing infrastructure development creates a substantial market for construction materials, increasing the exposure of domestic producers to potential future carbon pricing mechanisms or green procurement requirements.

3. ESG Disclosure Signals

- **CSRD Engineering & Construction Sector Report 2026 Highlights Disclosure Trends:** A recent report analyzing 43 engineering and construction companies reveals that 86% have formal GHG targets, and 42% have SBTi validated targets. This indicates a growing pressure for robust carbon accounting and disclosure within the sector, driven by EU ESG regulations and investor scrutiny. For construction clients with international operations or supply chains, aligning with these disclosure expectations is becoming a competitive necessity. (Source: KEY ESG, April 8, 2026).

4. Embodied Carbon Spotlight

Scenario: Structural Frame of a Hypothetical Commercial Building (approx. 5,000 m²) in Kazakhstan

Due to current tool limitations, specific tCO2e figures and percentage shares from an automated LCA cannot be provided for this briefing. However, based on typical industry data, we can illustrate the potential hotspots and savings.

Baseline Material Assumptions (Indicative Cradle-to-Gate Carbon Factors):

- Structural Steel (Basic Oxygen Furnace, BOF): ~2.0 kgCO2e/kg
- Concrete C30/37: ~250 kgCO2e/m³
- Steel Rebar: ~1.5 kgCO2e/kg

For a hypothetical building using approximately 100 tonnes of structural steel, 800 m³ of concrete, and 15 tonnes of rebar, the embodied carbon hotspots would primarily be the structural steel (due to high carbon intensity per kg) and the concrete (due to its high volume and cement content).

Hotspots:

- **Structural Steel:** Conventional BOF steel is highly carbon-intensive. This is a primary target for CBAM if exported to the EU or for domestic projects seeking lower carbon footprints.
- **Concrete:** The cement content in concrete is a major contributor to embodied carbon.

Circular Economy Recommendations (Qualitative Savings):

Material	Lower-carbon switch	Potential Saving (Qualitative)
Structural Steel	Switch to Electric Arc Furnace (EAF) steel with high recycled content (~0.5 kgCO2e/kg)	Significant reduction in embodied carbon (up to 75% for steel component) and reduced CBAM liability.

Material	Lower-carbon switch	Potential Saving (Qualitative)
Concrete C30/37	Utilize concrete mixes with high Supplementary Cementitious Materials (SCM) content (e.g., GGBS, fly ash)	Substantial reduction in cement-related emissions, contributing to overall project decarbonization.
Steel Rebar	Specify rebar produced via EAF routes	Lower embodied carbon compared to traditional blast furnace rebar, aligning with low-carbon objectives.
General Design	Optimize structural design to reduce material quantities; design for deconstruction/reuse	Reduces overall material demand and facilitates future circularity.

Embodied Carbon Trends

The rules for steel and concrete are fundamentally changing in 2026:

- **Procurement Shifts:** Demand for greener materials is increasingly shaping EU and UK procurement, moving from encouragement to expectation. Carbon data is now essential, not optional, with procurement frameworks rewarding transparency and low-emissions performance.
- **Carbon Disclosure as Baseline:** Carbon accounting is becoming a core business requirement for market competitiveness. Alignment of standards (e.g., ResponsibleSteel) is influencing procurement criteria and access to finance. Scrutiny is intensifying on the credibility and governance of accounting approaches, with attention on whether claims are linked to actual emissions reductions.
- **Performance-Based Concrete Standards:** The focus is shifting from rigid ingredient lists to performance-based standards. Digital tools and AI-enabled mix optimization are accelerating the adoption of low-carbon concrete by prioritizing strength, durability, and performance, opening doors for innovative new blends and materials.
- **Green Hydrogen Steel Scaling:** Green hydrogen-based Direct Reduced Iron (DRI) steel production is moving beyond pilot projects towards commercialization, offering a credible pathway to near-zero primary steel. New supply agreements and continued progress signal growing commercial confidence.
- **Investor Scrutiny:** Private developers, building owners, and corporate builders are integrating embodied carbon into material choices, driven by investor demands and the need to future-proof assets. Data centers are leading this trend, with their scale and long horizons driving early engagement with low-carbon concrete suppliers. (Source: The Climate Group, January 28, 2026).

Recommended Actions

1. **Prioritize Emissions Monitoring and Reporting for CBAM:** Immediately implement robust systems for measuring, calculating, and verifying actual carbon emissions for all aluminum, iron, and steel products destined for the EU. This is the single most effective way to halve CBAM liabilities and avoid punitive default charges.
2. **Integrate Kazakhstan's New Construction Code:** Review and update internal accounting policies, material procurement standards, and cost control procedures to align with the new Construction Code effective July 1, 2026. Ensure all project teams are aware of the new material consumption and costing requirements.
3. **Explore Low-Carbon Material Sourcing:** Actively seek suppliers of lower-carbon structural steel (e.g., EAF with high recycled content) and concrete (e.g., high SCM content). Begin requesting Environmental Product

Declarations (EPDs) for key materials to gather verifiable carbon data, which will be crucial for CBAM compliance, green building certifications, and future-proofing.

4. Assess Supply Chain Carbon Footprint: For companies involved in export to the EU or with European supply chains, begin mapping and assessing the embodied carbon footprint of key construction materials and components. This proactive step will identify hotspots and opportunities for decarbonization ahead of stricter regulations or client demands.

5. Advocate for Domestic Carbon Pricing Alignment: While a long-term goal, supporting the government's efforts to strengthen Kazakhstan's domestic emissions trading system could eventually reduce CBAM-related costs for exporters by allowing domestic carbon payments to be offset against EU charges.

Sources

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- <https://astanatimes.com/2026/02/kazakhstans-construction-sector-attracts-over-7-billion-in-investment>

Generated by the ESG Intelligence Agent on August 03, 2026. Embodied-carbon figures are indicative cradle-to-gate (A1-A3) estimates — verify with product-specific EPDs before formal reporting. Automated LCA tools were unavailable for this briefing.